# Valid is not explanatory

## What a proof about a cognitive model does and does not show, and an audit of CogLean's results

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October 2026

# Valid is not explanatory

- Lean guarantees **validity**: every theorem follows from its premises, with nothing hidden.
- It cannot say whether the premises **already contain** the phenomenon.
- A theorem "the model shows a bowed serial-position curve" is worth little if the model's definition is a bowed schedule.
- So two standards:
  - *formal*: does it follow? (Lean)
  - *explanatory*: how much of the behaviour is in the premises, and how much emerges? (scientific judgment)

# Three levels

1. **A phenomenon**: a pattern observed in people (errors bowed across a list).
2. **A predicate**: a precise statement of the pattern, in Lean. It asserts nothing; it is a target.
3. **A theorem**: a mechanism, under stated conditions, produces behaviour satisfying the predicate.

A predicate in CogLib's `Phenomena` is not an axiom. It becomes a result only when a mechanism is proved to satisfy it.

# Derived, or conditional?

- A theorem with a phenomenon among its *hypotheses* is a relation between phenomena: "if recency, then . . . ". Valid, useful, but not a prediction of any model.
- It becomes a prediction when a model *discharges* the hypothesis by proving it.
- The catalog now marks this mechanically: **Given**, the laws a theorem assumes.

| Theorem | Given | Status |
|---|---|---|
| confident errors from laws | determinacy, certainty, stability | conditional |
| confident misnaming (EPAM) | — (discharged from EPAM's cells) | derived |
| flat serial curve | certainty, stability | conditional |

32 of 565 theorems are conditional on laws.

# Explanatory distance

How far is the result from its premises?

| Distance | Meaning | Example |
|---|---|---|
| Definitional | a restatement | hierarchical network: cost is distance, so a distance effect |
| Close | the pattern is largely in the mechanism | selective modification's conjunction effect |
| Derived | follows from premises that don't mention it | the no-carry bug: deletion + impasse + critics |
| Compositional | needs two or more mechanisms | confident errors: recognition + associations |

The question to ask of every result: *how much of the behaviour is in the premises, and how much emerges?*

# Case 1: the serial-position curve

**Before**

- The model was a schedule: item $i$ learned on trial $2\times$ (distance from an end).
- The bow was in the definition. Valid; close to definitional.

**Now**

- A rule, over *any* anchors: nearer an anchor first (EPAM's M3).
- Proved: nearer is learned earlier, for any anchors.
- Ends as anchors $\Rightarrow$ the serial-position effect.
- A distinctive item as an anchor $\Rightarrow$ the **isolation effect**.

Still close, but one mechanism now answers to two phenomena.

# Case 2: the same deficit, two control regimes

Same operators, same missing carry operator; only the control differs.

| | no carry needed | a carry needed |
|---|---|---|
| Decision cycle with repair | exact | confident wrong (no-carry bug) |
| Means-ends analysis (GPS) | exact | no answer |

- **Compositional**: knowledge $\times$ control regime.
- **Discriminating**: the regimes differ in observable behaviour, with knowledge held constant.
- Proved for every problem.

# What the audit found

- **Strongest**: compositional results: confident errors; the arithmetic view (over-extension gone, typicality kept); the two regimes.
- **Derived**: the no-carry bug; GPS's units-first order; the isolation effect; DOA's counting error share; EPAM's oscillation.
- **Close**: serial position; the conjunction effect; DOA's closed forms.
- **Definitional**: the hierarchical network's costs; properties of algorithms (soundness, merges), which are library facts, not explanations.
- **Premises still unexplained**: sharp associations (assumed in every confident-errors result); prior associations; which items are anchors.

A judgment, written down (`notes/EXPLANATORY_AUDIT.md`); its weakest results are the targets.

# Conventions, not a checker

- Phenomena are **targets**, not premises.
- A theorem with a law among its hypotheses is **named and flagged** as conditional.
- Docstrings say which premises are **postulates** and which are **conditions**.
- Properties of algorithms are not presented as explanations.
- When a result is close to its definitions, say so, and look for a **second consequence** of the same mechanism.

Why not a checker? "Who checks the checker?" Lean already shows every premise; what remains is judgment, and the right place for it is in the open.

# Next

- The audit's weakest results: a process for semantic search, so distance effects are derived, not defined; sharp associations derived from learning; anchors derived from features.
- **A composition designed around this principle**: practising while sleepy. Fatigue raises errors; learning retains some of them; they persist when alert, unless feedback corrects them first. Stated with each premise marked as postulate, condition, or empirical premise awaiting explanation, and with the boundary cases as theorems too.